Execution Order of SQL Queries - Detailed Guide

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1. Introduction

Understanding the logical execution order of SQL queries is one of the most important concepts for mastering SQL. Although we write SQL queries in a specific order, the database engine processes them in a different logical sequence internally.

2. Written Order vs Logical Execution Order

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SQL Written Order:
SELECT → FROM → JOIN → WHERE → GROUP BY → HAVING → ORDER BY → LIMIT

Logical Execution Order:
FROM → JOIN → WHERE → GROUP BY → HAVING → SELECT → DISTINCT → ORDER BY → LIMIT/OFFSET
```

3. FROM Clause

The FROM clause identifies the base table(s) from which data is retrieved. This is the starting point of query execution.

4. JOIN Clause

JOIN combines rows from multiple tables based on a related column between them. All joins are logically processed before filtering occurs.

5. WHERE Clause

WHERE filters rows before any grouping or aggregation happens. Only rows that satisfy the condition move forward in execution.

6. GROUP BY Clause

GROUP BY groups rows sharing the same values in specified columns. It is used together with aggregate functions like COUNT, SUM, AVG, etc.

7. HAVING Clause

HAVING filters groups after aggregation has been performed. Unlike WHERE, it works on aggregated data.

8. SELECT Clause

SELECT determines which columns or expressions appear in the final result. Column aliases are created at this stage.

9. DISTINCT Clause

DISTINCT removes duplicate rows from the result set after SELECT is processed.

10. ORDER BY Clause

ORDER BY sorts the final result set in ascending or descending order.

11. LIMIT / OFFSET Clause

LIMIT restricts the number of rows returned, and OFFSET skips a specified number of rows.

12. Example Query with Execution Flow

```
SELECT d.name, COUNT(*) AS employee_count
FROM employees e
JOIN departments d ON e.dept_id = d.id
WHERE e.salary > 50000
GROUP BY d.name
HAVING COUNT(*) > 5
ORDER BY employee_count DESC
LIMIT 10;


Execution Breakdown:
1. FROM employees
2. JOIN departments
3. WHERE salary > 50000
4. GROUP BY department name
5. HAVING COUNT(*) > 5
6. SELECT department name and count
```

```
7. ORDER BY count descending
8. LIMIT 10 rows
```

13. Common Interview Traps

- Using column aliases in WHERE clause (fails because SELECT runs later).
- Using aggregate functions inside WHERE (must use HAVING instead).
- Confusion between WHERE and HAVING conditions.
- Misunderstanding when DISTINCT is applied.

14. Conclusion

To master SQL, always remember that SQL is written in one order but logically executed in another. Understanding this execution pipeline helps you write optimized queries, avoid common mistakes, and confidently answer interview questions.